

TCP/IP

1. What does TCP/IP stand for?

- A) Transmission Control Protocol / Internet Protocol
- B) Transmission Central Protocol / Internet Protocol
- C) Transfer Control Protocol / Internet Provider
- D) Transmission Control Provider / Internet Protocol

Answer: A) Transmission Control Protocol / Internet Protocol

2. How many layers are there in the standard TCP/IP model (original)?

- A) 7
- B) 4
- C) 5
- D) 6

Answer: B) 4

3. Which layer of the TCP/IP model corresponds to the OSI Physical and Data Link layers?

- A) Application Layer
- B) Transport Layer
- C) Internet Layer
- D) Network Access (or Link) Layer

Answer: D) Network Access (or Link) Layer

4. Which protocol is used for reliable, connection-oriented data transfer?

- A) UDP
- B) IP
- C) TCP
- D) ICMP

Answer: C) TCP

5. Which protocol is connectionless and unreliable?

- A) TCP
- B) UDP
- C) FTP
- D) SMTP

Answer: B) UDP

6. What is the length of an IPv4 address?

- A) 32 bits
- B) 64 bits
- C) 128 bits
- D) 48 bits

Answer: A) 32 bits

7. What is the length of an IPv6 address?

- A) 32 bits
- B) 64 bits
- C) 128 bits
- D) 256 bits

Answer: C) 128 bits

8. Which protocol is used to map an IP address to a MAC address?

- A) DHCP
- B) DNS
- C) ARP
- D) RARP

Answer: C) ARP

9. Which protocol is used to map a MAC address to an IP address?

- A) ARP
- B) RARP

- C) DNS
- D) ICMP

Answer: B) RARP

10. Port number 80 is reserved for:

- A) FTP
- B) HTTPS
- C) HTTP
- D) Telnet

Answer: C) HTTP

11. Which layer of the TCP/IP model handles routing of packets across networks?

- A) Application
- B) Transport
- C) Internet
- D) Network Access

Answer: C) Internet

12. Which protocol is used for sending emails?

- A) POP3
- B) IMAP
- C) SMTP
- D) HTTP

Answer: C) SMTP

13. Which protocol is used for retrieving emails from a server?

- A) SMTP
- B) SNMP
- C) POP3
- D) UDP

Answer: C) POP3

14. What does DNS stand for?

- A) Data Name Service
- B) Domain Name System
- C) Domain Network Service
- D) Data Network System

Answer: B) Domain Name System

15. The combination of an IP address and a port number is called a:

- A) Socket
- B) Link
- C) Node
- D) Session

Answer: A) Socket

16. Which flag is set in the first step of the TCP 3-way handshake?

- A) ACK
- B) FIN
- C) SYN
- D) RST

Answer: C) SYN

17. What is the standard port number for HTTPS?

- A) 80
- B) 8080
- C) 443
- D) 21

Answer: C) 443

18. Which protocol allows dynamic IP assignment to clients?

- A) DNS

B) DHCP

C) ARP

D) NAT

Answer: B) DHCP

19. Ping command uses which protocol?

A) TCP

B) UDP

C) ICMP

D) IGMP

Answer: C) ICMP

20. Which layer provides end-to-end communication and error recovery?

A) Internet Layer

B) Network Access Layer

C) Transport Layer

D) Application Layer

Answer: C) Transport Layer

21. FTP uses which ports?

A) 20 and 21

B) 23 and 25

C) 80 and 443

D) 110 and 143

Answer: A) 20 and 21

22. What acts as a firewall in a network?

A) To speed up the network

B) To stop unauthorized access

C) To route packets

D) To assign IP addresses

Answer: B) To stop unauthorized access

23. Which protocol is used for secure remote login?

A) Telnet

B) SSH

C) FTP

D) HTTP

Answer: B) SSH

24. In TCP/IP, the "Host-to-Host" layer is equivalent to which OSI layer?

A) Network

B) Transport

C) Session

D) Presentation

Answer: B) Transport

25. The size of the TCP header is generally:

A) 20 bytes

B) 32 bytes

C) 64 bytes

D) 8 bytes

Answer: A) 20 bytes (minimum)

26. Which of the following is a Loopback address?

A) 192.168.1.1

B) 127.0.0.1

C) 10.0.0.1

D) 255.255.255.0

Answer: B) 127.0.0.1

27. What is the maximum size of an IP packet?

A) 65535 bytes

B) 1500 bytes

C) 32768 bytes

D) 1024 bytes

Answer: A) 65535 bytes

28. Which protocol is used to manage and monitor network devices?

A) SMTP

B) SNMP

C) FTP

D) HTTP

Answer: B) SNMP

29. The process of dividing a large packet into smaller fragments is handled by:

A) TCP

B) IP

C) UDP

D) ARP

Answer: B) IP

30. Flow control in TCP is handled using:

A) Sliding Window

B) Stop and Wait

C) Go-Back-N

D) Checksum

Answer: A) Sliding Window

31. Which class of IP address allows the most number of hosts?

A) Class A

B) Class B

C) Class C

D) Class D

Answer: A) Class A

32. The subnet mask 255.255.255.0 corresponds to a CIDR notation of:

A) /8

B) /16

C) /24

D) /32

Answer: C) /24

33. Which protocol is used for file transfer without authentication (unreliable)?

A) FTP

B) TFTP

C) SFTP

D) HTTP

Answer: B) TFTP

34. A MAC address is how many bits long?

A) 32

B) 48

C) 64

D) 128

Answer: B) 48

35. Which layer defines the hardware addressing (MAC)?

A) Internet Layer

B) Transport Layer

C) Network Access Layer

D) Application Layer

Answer: C) Network Access Layer

36. Which TCP flag indicates that there is no more data from the sender?

- A) SYN
- B) PSH
- C) FIN
- D) URG

Answer: C) FIN

37. The 'Time to Live' (TTL) field is found in the header of:

- A) TCP
- B) UDP
- C) IP
- D) Ethernet

Answer: C) IP

38. Which protocol provides multicasting support?

- A) ICMP
- B) IGMP
- C) SNMP
- D) SMTP

Answer: B) IGMP

39. In a Class C IP address, how many bits are used for the network ID?

- A) 8
- B) 16
- C) 24
- D) 32

Answer: C) 24

40. What is the default port for Telnet?

- A) 21
- B) 22

C) 23

D) 25

Answer: C) 23

41. Which of the following is a Private IP address?

A) 192.168.1.1

B) 172.15.0.1

C) 11.0.0.1

D) 8.8.8.8

Answer: A) 192.168.1.1

42. Checksum in TCP/IP is used for:

A) Routing

B) Flow Control

C) Error Detection

D) Addressing

Answer: C) Error Detection

43. Which protocol handles the resolution of logical addresses to physical addresses?

A) ARP

B) RARP

C) DNS

D) DHCP

Answer: A) ARP

44. Which layer of TCP/IP interacts directly with the user software?

A) Application

B) Transport

C) Internet

D) Link

Answer: A) Application

45. Sequence numbers are used by which protocol?

- A) UDP
- B) IP
- C) TCP
- D) ICMP

Answer: C) TCP

46. What does NAT stand for?

- A) Network Address Translation
- B) Network Access Tool
- C) Node Allocation Table
- D) Native Address Transport

Answer: A) Network Address Translation

47. IPv6 uses which notation for addresses?

- A) Dotted Decimal
- B) Hexadecimal Colon
- C) Binary
- D) Octal

Answer: B) Hexadecimal Colon

48. Which protocol ensures data is delivered in the correct order?

- A) IP
- B) UDP
- C) TCP
- D) Ethernet

Answer: C) TCP

49. Port 53 is used by:

- A) DHCP

- B) DNS
- C) SMTP
- D) FTP

Answer: B) DNS

50. Which command displays the routing table on a computer?

- A) ping
- B) ipconfig
- C) route print / netstat -r
- D) tracert

Answer: C) route print / netstat -r

51. The "Three-Way Handshake" involves:

- A) SYN, SYN-ACK, ACK
- B) ACK, SYN, FIN
- C) SYN, ACK, SYN
- D) FIN, ACK, SYN

Answer: A) SYN, SYN-ACK, ACK

52. UDP is often used for:

- A) File Transfer
- B) Email
- C) Streaming Media / VoIP
- D) Web Browsing

Answer: C) Streaming Media / VoIP

53. Which field in the IP header prevents a packet from looping endlessly?

- A) Checksum
- B) Fragment Offset
- C) TTL (Time To Live)
- D) Version

Answer: C) TTL (Time To Live)

54. MIME stands for:

- A) Multi-purpose Internet Mail Extensions
- B) Multiple Internet Mail Exchange
- C) Mail Internet Message Extension
- D) Multi-media Internet Mail Environment

Answer: A) Multi-purpose Internet Mail Extensions

55. Which protocol allows a user to access a mailbox on a server and manipulate messages as if they were local?

- A) POP3
- B) IMAP
- C) SMTP
- D) FTP

Answer: B) IMAP

56. In IPv4, Class D addresses are used for:

- A) Unicast
- B) Multicast
- C) Broadcast
- D) Research

Answer: B) Multicast

57. 127.0.0.0 to 127.255.255.255 is reserved for:

- A) Private networks
- B) Multicast
- C) Loopback testing
- D) Broadcast

Answer: C) Loopback testing

58. Which of the following is NOT a layer in the TCP/IP model?

- A) Internet
- B) Session
- C) Application
- D) Transport

Answer: B) Session

59. The OSI model has 7 layers. The TCP/IP Application layer covers which OSI layers?

- A) Application only
- B) Application and Presentation
- C) Application, Presentation, and Session
- D) Session and Transport

Answer: C) Application, Presentation, and Session

60. Which protocol enables a diskless workstation to boot up and find its IP address?

- A) ARP
- B) RARP (or BOOTP/DHCP)
- C) ICMP
- D) SMTP

Answer: B) RARP (or BOOTP/DHCP)

61. A broadcast address generally ends with:

- A) 0
- B) 255
- C) 1
- D) 100

Answer: B) 255 (in a standard /24 network)

62. Which command is used to trace the path a packet takes to a destination?

- A) ping
- B) traceroute / tracert

- C) ipconfig
- D) nslookup

Answer: B) traceroute / tracert

63. The header length of an IPv4 packet is typically:

- A) 20 bytes
- B) 40 bytes
- C) 64 bytes
- D) 128 bytes

Answer: A) 20 bytes

64. What is the port number for SMTP?

- A) 21
- B) 23
- C) 25
- D) 80

Answer: C) 25

65. Which TCP/IP protocol is used to synchronize clocks on computers?

- A) NTP
- B) SMTP
- C) SNMP
- D) POP

Answer: A) NTP

66. "Best Effort Delivery" describes which protocol?

- A) TCP
- B) IP
- C) HTTP
- D) FTP

Answer: B) IP

67. Which of the following is a Link State Routing Protocol?

- A) RIP
- B) OSPF
- C) IGRP
- D) EIGRP

Answer: B) OSPF

68. RIP (Routing Information Protocol) is a:

- A) Distance Vector Protocol
- B) Link State Protocol
- C) Hybrid Protocol
- D) Exterior Gateway Protocol

Answer: A) Distance Vector Protocol

69. Which device works at the Network Access / Data Link layer?

- A) Router
- B) Switch
- C) Hub
- D) Repeater

Answer: B) Switch

70. Which device works at the Internet / Network layer?

- A) Switch
- B) Router
- C) Hub
- D) Bridge

Answer: B) Router

71. DHCP uses which transport protocol?

- A) TCP

- B) UDP
- C) IP
- D) ICMP

Answer: B) UDP

72. The term "Octet" in an IP address refers to:

- A) 4 bits
- B) 8 bits
- C) 16 bits
- D) 32 bits

Answer: B) 8 bits

73. How many ports are available in TCP/UDP?

- A) 1024
- B) 32768
- C) 65535
- D) Unlimited

Answer: C) 65535

74. Ports 0 to 1023 are known as:

- A) Well-known ports
- B) Registered ports
- C) Dynamic ports
- D) Private ports

Answer: A) Well-known ports

75. Which protocol is used to secure HTTP connections?

- A) SSH
- B) SSL/TLS
- C) IPSec
- D) SFTP

Answer: B) SSL/TLS

76. What is the loopback address in IPv6?

- A) ::1
- B) 127.0.0.1
- C) ::
- D) 2001::1

Answer: A) ::1

77. An IPv4 address consists of how many octets?

- A) 2
- B) 4
- C) 6
- D) 8

Answer: B) 4

78. The "Protocol" field in the IP header identifies:

- A) The source IP
- B) The destination IP
- C) The transport layer protocol (TCP/UDP) carrying the data
- D) The packet length

Answer: C) The transport layer protocol (TCP/UDP) carrying the data

79. In TCP, what does the RST flag signify?

- A) Reset the connection
- B) Request synchronization
- C) Push data
- D) Finish connection

Answer: A) Reset the connection

80. What is a "Socket Address"?

- A) IP Address + MAC Address
- B) IP Address + Port Number
- C) MAC Address + Port Number
- D) URL + IP Address

Answer: B) IP Address + Port Number

81. Which protocol is used for chat/instant messaging (historically)?

- A) IRC
- B) FTP
- C) SNMP
- D) NTP

Answer: A) IRC

82. Which version of IP provides built-in security (IPSec)?

- A) IPv4
- B) IPv5
- C) IPv6
- D) IPv4s

Answer: C) IPv6

83. The maximum size of the data field in an Ethernet frame (MTU) is typically:

- A) 1000 bytes
- B) 1500 bytes
- C) 2048 bytes
- D) 4096 bytes

Answer: B) 1500 bytes

84. If a destination IP is not in the local subnet, where is the packet sent?

- A) To the switch
- B) To the Default Gateway (Router)
- C) Dropped

D) Broadcasted

Answer: B) To the Default Gateway (Router)

85. Which command resolves an IP address to a hostname?

A) ping -a

B) ipconfig

C) netstat

D) arp -a

Answer: A) ping -a (or nslookup)

86. Which protocol layer is responsible for logical addressing?

A) Link Layer

B) Internet Layer

C) Transport Layer

D) Application Layer

Answer: B) Internet Layer

87. Which protocol is used to copy files securely?

A) FTP

B) TFTP

C) SCP

D) Telnet

Answer: C) SCP

88. What happens when the TTL of a packet reaches 0?

A) It is delivered

B) It is discarded/dropped

C) It is returned to sender

D) It is retransmitted

Answer: B) It is discarded/dropped

89. A proxy server is often used to:

- A) Assign IP addresses
- B) Hide internal network details/Cache content
- C) Route packets
- D) Resolve names

Answer: B) Hide internal network details/Cache content

90. In the URL http://www.example.com, "http" represents the:

- A) Domain
- B) Protocol
- C) Path
- D) Server

Answer: B) Protocol

91. Which protocol is used for browsing the World Wide Web?

- A) FTP
- B) HTTP
- C) SMTP
- D) SNMP

Answer: B) HTTP

92. The "Window Size" field in a TCP segment tells the sender:

- A) How much data can be sent before an acknowledgment is required
- B) The size of the packet
- C) The size of the header
- D) The sequence number

Answer: A) How much data can be sent before an acknowledgment is required

93. TCP/IP was developed by:

- A) ISO
- B) DARPA (Department of Defense)

C) CERN

D) IEEE

Answer: B) DARPA

94. Which class of IP address is reserved for experimental purposes?

A) Class C

B) Class D

C) Class E

D) Class B

Answer: C) Class E

95. In a TCP connection, which state represents waiting for a connection request from any remote TCP?

A) ESTABLISHED

B) LISTEN

C) SYN-SENT

D) CLOSED

Answer: B) LISTEN

96. Which networking command shows active TCP connections and ports?

A) ipconfig

B) netstat

C) ping

D) tracert

Answer: B) netstat

97. An "Anycast" address in IPv6 identifies:

A) A single interface

B) A set of interfaces, but packet is delivered to the nearest one

C) All interfaces

D) A multicast group

Answer: B) A set of interfaces, but packet is delivered to the nearest one

98. Which protocol is used to send error messages indicating that a requested service is not available?

- A) TCP
- B) IP
- C) ICMP
- D) IGMP

Answer: C) ICMP

99. In sub-netting, borrowing bits from the host portion to the network portion:

- A) Increases the number of hosts
- B) Decreases the number of subnets
- C) Increases the number of subnets
- D) Has no effect

Answer: C) Increases the number of subnets

100. The default subnet mask for a Class A address is:

- A) 255.0.0.0
- B) 255.255.0.0
- C) 255.255.255.0
- D) 255.255.255.255

Answer: A) 255.0.0.0